BRILLIANTS' CONVENT



CLASS-XII

2024-25

SUMMER HOLIDAY HOMEWORK

Class XI

PHYSICS

1. Revise chapter Electric field and charges, Electrostatic potential and capacitance, Current electricity.
2. Solve all examples and back exercise of NCERT of chapters 1,2,3.
3. Write all frequently asked questions of chapter 1,2,3 from PDF given to you.
4. Write eight practicals in the practical file. Do not write readings of the practicals which you have not done in physics lab.

LIST OF PRACTICALS:

1. To determine resistivity of one/two wires by plotting a graph for potential difference versus current.
2. To find resistance of a given wire / standard resistor using meter bridge.
3. To verify the laws of combination (series) of resistances using a meter bridge.
4. To determine resistance of a galvanometer by half-deflection method and to find its figure of merit.
5. To find the focal length of a convex lens by plotting graphs between u and v.
6. To find the focal length of a convex mirror, using a convex lens.
7. To determine angle of minimum deviation for a given prism by plotting a graph between angle of incidence and angle of deviation.
8. To determine angle of minimum deviation for a given prism by plotting a graph between angle of incidence and angle of deviation.

Note: Write six activities in a separate folder as discussed in class.

LIST OF ACTIVITIES:

1. To assemble a household circuit comprising three bulbs, three (on/off) switches, a fuse and a power source.
2. To assemble the components of a given electrical circuit.
3. To draw the diagram of a given open circuit comprising at least a battery, resistor/rheostat, key, ammeter and voltmeter. Mark the components that are not connected in proper order and correct the circuit and also the circuit diagram.
4. To observe diffraction of light due to a thin slit.
5. To observe refraction and lateral deviation of a beam of light incident obliquely on a glass slab.
6. To study the nature and size of the image formed by a convex lens/concave mirror on a screen by using a candle and a screen for different distances of a candle from the lens/mirror.

Note:-Make one investigatory project preferably from your syllabus and prepare a file for the same

CHEMISTRY

1. Build your own element from the natural aspect from Andaman Nico bar Islands

   Directions: Build your own element (observing all forms of chemical nature and studies of natural elemental aspect in Islands). Use the following information:

   Name of the element: use your class roll number (confirm your roll number from Your class teacher)

   *Symbol, atomic number, mass number

   *Electronic configuration & Electronic structure

   - Isotopes / Allotropes (if any)
   - Position (group & period) of the element in the modern periodic table
   - History of the element
   - Importance of the element and its compounds in various fields
   - Uses of at least 10 important compounds of the element
   - Learning objectives: * Learn about various elements, importance of these
   - Elements in various fields and the structural aspects of these elements.
   - Rubrics:
   - Creativity: 5 marks Content: 5 marks Presentation: 5 marks
   - • Date of submission: 14th July 2024.

2. Prepare for Pre mid Exams.

BIOLOGY

1. 'The rate of appearance of new forms is linked to life-cycle or the life span'. Explain this statement in the light of natural selection theory of Darwin.

2. How does industrial melanism confirm the natural selection theory of Darwinism?

3. Differentiate between convergent & divergent evolution.

4. Draw a labeled diagram of DNA molecule.

5. Give an outline of steps involved in DNA fingerprinting.

6. In E.coli, 3 enzymes β galactosidase, permease & transacetylase began to be produced as soon as lactose was added. Explain why did the enzyme not form in the absence of lactose?

7. Does each of the 2 complementary strands of DNA carry the same biological message?

8. Differentiate between following

a) Repetitive DNA and satellite DNA
b) mRNA & tRNA
c) Template strand & coding strand

9. Explain pleiotropy with a suitable example.

10.A cross was made between a hybrid tall and red flowered plant (TtRr) with dwarf and white flowered plant (ttrr). What will be the genotypes of plants of f1 progency?

Note: Do al these questions in Biology notebook.

INNOVATIVE PROJECT:

1.Complete practical aspect of project work. Write introduction & theory.

2.Complete practical file

MATHS

1. Practice and revise all chapters taught in class till date.
2. Learn all the formulas that are there in certain chapters along with writing down on a paper.
3. Do the NCERT Exercises questions along with Miscellaneous exercise in your notebook.
4. Solve NCERT Exemplar questions in your notebook.
5. Make one investigatory project preferably from your syllabus and prepare a file for the same.

ECONOMICS

1. Revise the chapters covered in class.

2. Prepare project file as per the assigned roll numbers and topics

- Micro and small scale industries

  XII B - Roll No.1

  XII C –Roll No.-1

- Contemporary employment situation in India

  XII B - Roll No. 2

  XII C –Roll No.2

- Human development Index-

  XII B Roll No. 3

  XII C Roll No.3

- Cashless economy

  XII B - Roll No. 4

  XII C Roll No.-4

- G-20 Inclusive and action oriented

  XII B - Roll No. 5

  XII C Roll No.-5

- Sustainable development goals

  XII B - Roll No. 6

  XII C Roll No.-6

- Aatmanirbhar Bharat

  XII B - Roll No. 7

  XII C Roll No. 7

- Make In India

  XII B - Roll No. 8

  XII C Roll No. 8

- Digital India

  XIIB-Roll no. 9

  XII C- Roll no. 9

- Organic farming

  XII B- Roll no. 10

  XII C Roll no. 10

- Indian Economy on the Eve of Independence

  XII B Roll no. 11

  XII C Roll no. 11,12

ACCOUNTANCY

1. Revise the syllabus covered in the class.

2. Prepare a specific project of any company including its

   A. Segment reporting

   B. Cash flow Statement

BUSINESS STUDIES

1. Revise the syllabus covered in the class

2. Prepare the project allotted to you according to your Roll numbers

   - Business Environment (Roll no. 2, 4, 11)
   - Principles of management (Roll no. 1, 6, 8)
   - Marketing management (Roll no. 3, 5, 7)
   - Stock exchange (Roll no. 9, 10)

ENGLISH

1. INTRODUCTION

The project consists of 10 MARKS out of which, 5 MARKS will be allotted for the PROJECT FILE and the remaining 5 MARKS for the VIVA based on the file.

2. CONTENT OF THE PROJECT FILE:

The project file should include the following:

- Cover page, with the title of the project, school details and details of the student
- Certificate of Completion under the guidance of the teacher
- Objectives of the topic
- Essay/report should be written in 800-1000 words.
- Student reflections (the new learning experience/outcome achieved after completing the project)
- Photographs that capture positive learning experience of the students (collages/pics from various online sources) can be pasted.
- List of Resources/Bibliography (Last page of the project file)

3. INSTRUCTIONS:

- Listen to podcasts, documentaries, interviews etc. on the given topics.
- Do a thorough research on the topic assigned.
- Prepare a report/essay in about 800-1000 words describing the topic/issue/ giving your own opinion/ suggestions/measures/ viewpoints/its impact on people/your learning experience.
- The project should be neat, legible, with an emphasis on quality of content, accuracy of information, creative expression, proper sequencing and relevance as per the assigned topic.
- Use coloured practical sheets.
- Plagiarism is strictly prohibited.
  Note: Select any one of the given topics for your English Project.

1. CHILD LABOUR IN INDIA

"Child labour perpetuates poverty, unemployment, illiteracy, population growth and other social problems." Kailash Satyarthi.

You have studied the lesson 'Lost Spring' by Anees Jung. It deals with child labour in India. Based on your understanding of the lesson and the problem it mentions, make a project titled 'CHILD LABOUR IN INDIA'

Include the following sub-headings:

- Causes of child labour
- Effects on the social environment
- Laws on child labour
- National child labour policies & schemes
- Prevention and elimination of child labour
- Industries employing children- Make a case study (800-1000 words) on any 1 industry that employs child labour. Furnish all relevant details and statistics. Attach 3-4 pictures.

SAY NO TO WAR

"An eye for an eye only ends up making the whole world blind." Mahatma Gandhi

You have studied the lesson 'The Last Lesson' by Alphonse Daudet. It deals with war and the long-lasting effects it has on the society at large. Based on your understanding of the lesson and the problem it mentions, make a project titled 'SAY NO TO WAR'

Include the following sub-headings:

- Definition of war
- Effects on social, economic environment

- Cost, in terms of money and lives
- Important international treaties
- Prevention and elimination of wars
- Any 1 war – Make a case study (800 – 1000 words) and furnish all details and statistics. Attach 3-4 pictures.

2. Revise the syllabus done so far.

POLITICAL SCIENCE

Note: To be done in Political Science Notebook.

1. Who was Gorbachev? Write a short note on him.
2. What was the Soviet system?
3. Mention the year and the relevance of the breaking of the Berlin wall?
4. What role did the World Bank play in integrating the Soviet states?
5. As a result of shock therapy to which economic system, each state of the Soviet bloc was gradually to be absorbed?
6. Mention two characteristics of the Soviet political system.
7. Mark the states that emerged from the Soviet Union on a political map of the world.
8. Mention areas of diplomacy between India and USSR?
9. Bring out the difference between multipolar, bipolar and unipolar world.
10. Mention any three features that distinguish the Soviet economy from that of a capitalist country like the US. What is the 'Two Nation Theory' advanced by Muslim League?
11. What do you understand by 'instrument of accession'?
12. What two challenges were faced by India at the time of her independence?
13. What was meant by princely states? How many princely states were there in India at the time of independence?
14. Mention the role of Sardar Vallabhbhai Patel towards India's integration.
15. Describe the outcomes of States Reorganisation Commission appointed in 1953

PROJECT

Some suggested topics are:

1. Disintegration of USSR with special focus on Gorbachev (**Roll no 1-3**)
2. Arab Spring, ASEAN, European Union, BRICS, SAARC (**Roll no 4-6**)
3. Cover the negative as well as positive aspects of relationship between (**Roll no 7-9**)

   Relationship between India and Russia

   Relationship between India and China

   Relationship between India and Israel
4. Partition of India-Theory behind it and its legacy (**Roll no 10-12**)

Extra topics-If someone wants to opt for.

1. Comparison between NITI AAYOG and Planning Commission- and their contribution, five-year plan in India's Development

2. UN Agencies – UNICEF, UNESCO, WHO, India's Nuclear Policy

General Instruction:

1. It should be a handwritten project on an A-4 size sheets.

2. It should be well researched and pictorial.

3. The project must have table of content.

4. It must include relevant information, facts and figures

5. It should cover following parameters (title/cover, page/, acknowledgement/ bibliography/ headings/sub headings)

CS & IP

Q1. Do the following Python programs practically and make a single pdf file.
The format of the file is as follows:

- Program heading (Question)
- Code of the program question
- Screenshot of the output

Questions:

1. WAP to compute x n of given two integers x and n.
2. WAP for calculating simple interest.

3. WAP to accept a number from the user and display whether it is an even number or odd number.
4. WAP to accept percentage of a student and display its grade accordingly.
5. WAP to print Fibonacci series up to certain limit.
6. WAP to display prime numbers up to a certain limit.
7. WAP to accept a number, find and display whether it's a Armstrong number or not.
8. WAP to accept a number and find out whether it is a perfect number or not.
9. WAP to print the sum of the series 1+x1/1!+x2/2!+…….xn/(n)!- exponential series.
10. WAP to print the following pattern:

1
1 2
1 2 3

11 WAP to accept a string and display whether it is a palindrome.
12. WAP that counts the number of alphabets and digits, uppercase letters, lowercase letter, spaces and other characters in the string entered.
13. WAP to accept a string (a sentence) and returns a string having first letter of each word in capital letter.
14. WAP to remove all odd numbers from the given list.
15. WAP to display second largest element of a given list.
16. WAP to display cumulative elements of a given list.
17. WAP to display frequencies of all the elements of a list.
18. WAP in Python to display those strings which are string with 'A' of given list.
19. WAP in Python to find and display the sum of all the values which are ending with 3 from a list.
20. WAP to shift the negative number to left and the positive numbers to right.
21. WAP to swap the content with next value divisible by 7.
22. WAP to accept values from user and create a tuple.
23. Write a program to input total number of sections and stream name in 11th class and display all information on the output screen.
24. Write a Python program to input names of 'n' countries and their capital and currency, store it in a dictionary and display in tabular form. Also search and display for a particular country.

Informatics Practices

Q1. Do the following Python programs practically and make a single pdf file.

- The format of the file is as follows:
- Program heading (Question)
- Code of the program question
- Screenshot of the output

Questions

1. Create a pandas series from a dictionary of values and an array.
2. Given a Series, print all the elements that are above the 75th percentile.
3. Create a data frame for examination result and display row labels, column labels data types of each column and the dimensions
4. Filter out rows based on different criteria such as duplicate rows.
5. Locate the 3 largest values in a data frame.
6. Subtract the mean of a row from each element of the row in a Data Frame.
7. Replace all negative values in a data frame with a 0.
8. Importing and exporting data between pandas and CSV file.
9. Importing and exporting data between pandas and MySQL database.
10. Given the school result data, analyse the performance of the students on different parameters, e.g subject wise or class wise.
11. For the Data frames created above, analyze and plot appropriate charts with title and legend.
12. Program to plot frequency of marks using Line chart.
13 Program to draw two lines along with proper titles and legends.

HINDI

प्रश्न 1- प्रत्येक छात्र /छात्रा अपने अनुक्रमांक के अनुसार दिए गए विषयों से संबंधित एक परियोजना कार्य तैयार करेंगे, जिसमें 900 से 1000 शब्द होंगे। परियोजना कार्य बहुत ही स्वच्छ और त्रुटि रहित होना चाहिए।

यही परियोजना कार्य आपके वार्षिक प्रयोगात्मक परीक्षा में प्रस्तुत किया जाएगा,इसलिए कार्य ध्यानपूर्वक व स्वच्छता पूर्ण तरीके से किया जाए।

1-अनुक्रमांक-1,7

कंप्यूटर एक जादुई पिटारा/सुविधाओं का भंडार: कंप्यूटर

2- अनुक्रमांक- 2,8

बच्चों की शिक्षा में माता पिता की भूमिका

3- अनुक्रमांक- 3,9

जीना मुश्किल करती महंगाई

4- अनुक्रमांक - 4,10

विश्व की शान बनता भारत

5- अनुक्रमांक- 5,11

मानव मन को संबोधित करते विज्ञापन/विज्ञापनों का पड़ता न नई पीढ़ी पर प्रभाव

6- अनुक्रमांक- 6,12

बच्चों के जीवन में कोडिंग से होने वाले बदलाव

पहला पृष्ठ- विद्यार्थी का नाम, कक्षा, वर्ग,अनुक्रमांक ,विद्यालय का नाम एवं विद्यालय का प्रतीक चिह्न (लोगो)

दूसरा पृष्ठ-आभार ज्ञापन

तीसरा पृष्ठ- प्रमाण-पत्र

चौथा पृष्ठ- अनुक्रमणिका

पॉचवा पृष्ठ- भूमिका/प्रस्तावना

छठे पृष्ठ से - संपूर्ण चित्राकर्षक परियोजना

सातवां पृष्ठ- संदर्भ- ग्रंथ

अंतिम पृष्ठ- हस्ताक्षर

प्रश्न:2 विशेष परीक्षा पाठ्यक्रम के समस्त पाठों का ध्यानपूर्वक पुनः अभ्यास कीजिए।

PHYSICAL EDUCATION

Q1. Make the project files as per the topics discussed in the class.

Q2. Revise the entire syllabus covered so far in the class.